

Candy Lottery

Time limit: 1.00 s

Memory limit: 512 MB

There are n children, and each of them independently gets a random integer number of candies between 1 and k .

What is the expected maximum number of candies a child gets?

Input

The only input line contains two integers n and k .

Output

Print the expected number rounded to six decimal places (rounding half to even).

Constraints

- $1 \leq n \leq 100$
- $1 \leq k \leq 100$

Example

Input	Output
2 3	2.444444